

STAT 3113 – Applied Engineering Statistics

Group Project Report Requirements

Final Report: Submit a final report by **12/17 at 11:59pm** on Canvas. Use $\alpha = 0.05$ for all testing. The final report should include the following parts:

- a. Introduction: Clearly state the objective of the experiment, the factors involved, the factor levels used, define the design, the number of replicates, and the response variable of interest.
- b. Methodology: Discuss how the treatment combinations were randomly applied to the experimental units and how the order of collecting data was randomized. Describe how you conducted the experiment and gathered the data. A list of raw data should be included in an appendix.
- c. Analysis and Interpretation of Results: Run all data analysis including checking conditions and running all appropriate hypothesis tests. Be clear on the order of the process. State all null and alternative hypotheses. Interpret each ANOVA test necessary even if an underlying condition is violated.
- d. Conclusions: Interpret the results and state your overall conclusions. Also, discuss any issues you encountered and how you might change the experiment to fix these issues if you redid it.
- e. Appendices: Include the raw data in one appendix. In another, include a picture of the group members conducting the experiment.

You should not provide pages and pages of computer output. Copy and paste only the relevant results. If superfluous results and excessive computer output are attached, up to 10 points would be subtracted from your total project score.

THERE WILL BE EXAMPLE PROJECT REPORTS POSTED. READ THOSE TO SEE WHAT IS EXPECTED.